

1 Correctness-adjusted circuit cost

To estimate whether test-set hunting concealed rare failures, we evaluated 42 historical accepted circuits on a common corpus of exactly 50,000 valid `secp256k1` point-addition input pairs. The sampled history contains accepted submissions 1, 11, 21, and so on, together with submissions 401 and 405, the last accepted commit in the repository. The pairs were generated pseudorandomly from the fixed seed `ecdsa-fail-qxt-history-v1`, independently of each circuit’s operation stream. Every commit therefore received the same inputs. This paired design makes the comparison reproducible and prevents any circuit from benefiting from a specially hunted Fiat–Shamir test set.

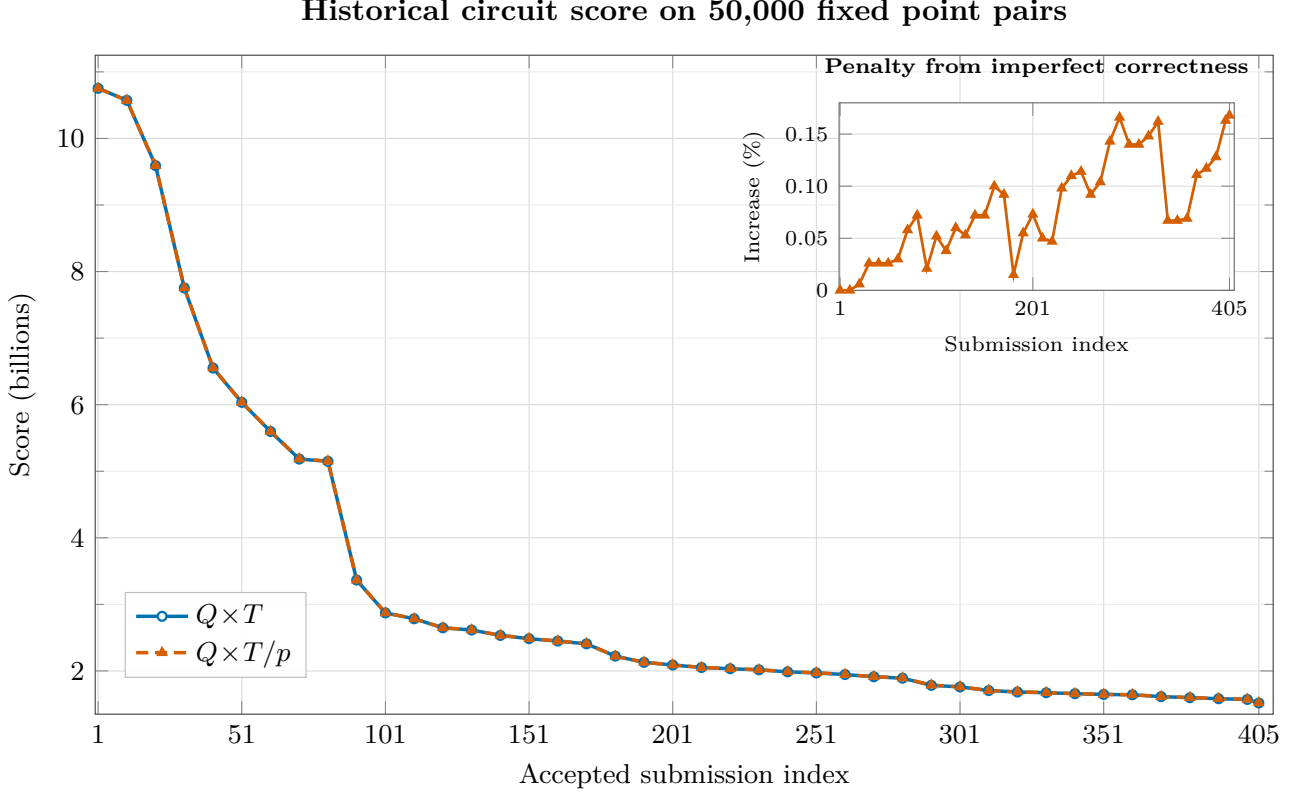


Figure 1: Raw and correctness-adjusted circuit scores across the sampled accepted submissions. The main panel plots $Q \times T$ and $Q \times T/p$; lower values are better. Because the two series are close at the full score scale, the inset reports the relative increase $100(1/p - 1)$ in percent.

For each circuit, Q is the peak number of live qubits, T is the average executed Toffoli count on the shared corpus, and p is the observed fraction of inputs that produced the expected classical affine output. We report the raw score and its reliability-adjusted counterpart as

$$S_{\text{raw}} = QT, \quad S_{\text{adjusted}} = \frac{QT}{p}.$$

The factor $1/p$ penalizes unreliable circuits and can be interpreted as the expected number of independent attempts per successful result when failures are detectable. Here, “correct” refers specifically to the classical output; phase and ancilla-cleanliness diagnostics were recorded separately and were not folded into p .

Across the sampled commits, correctness ranged from 100% to 99.832%, with a median of 99.928%. Consequently, the adjustment was small: its maximum relative increase was only 0.168%. For the final commit, the raw score was 1.518693 billion and the adjusted score was 1.521249 billion. The raw and adjusted curves therefore nearly overlap in Figure 1, although their separation is visible in the inset. No ranking reversal occurred among the 42 sampled circuits: every circuit that ranked better under $Q \times T$ also ranked better under $Q \times T/p$. This agreement is a property of these measurements, not of the metric in general; a circuit with a slightly lower raw score could rank worse after adjustment if its correctness probability were sufficiently lower. From the first to the last sampled commit, the raw score improved by a factor of 7.08, while the adjusted score improved by a factor of 7.07.